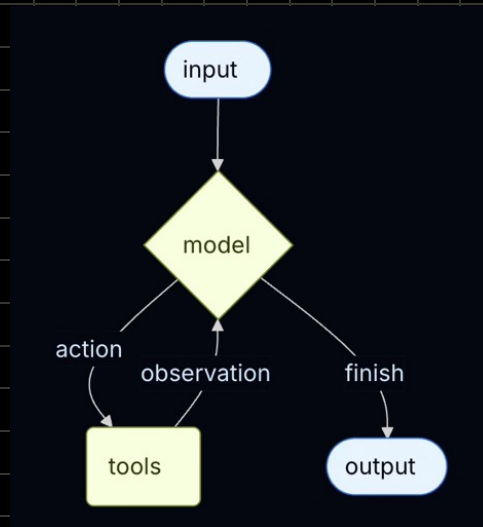


Langchain Documentation

Agents

Agents combine LLM with tools to create system that reasons about the tasks, decides which tool to use and iteratively work towards reaching goals.

The LLM agent runs tools in loop until it achieves the goal or iteration limit is reached.



Core Components of Agents

- 1) Model
- 2.) Tools
- 3) Systems Prompt
- 4.) Memory
- 5) Messages.
- (6) I/O

Models

The model is the reasoning engine of the agent.

Models drive the agent's decision making process, determining which tool to use, how to interpret results and when to provide answer

① Initialize The model.

```
model = init_chat_model ( )
```

- model name
- temperature
- max_tokens
- timeout
- max_retries

② model Invocation

- invoke
- stream
- Batch

Invoke returns a single output after having a complete output generated. But stream will return multiple objects where each will have portion of the output & gathered into full output via summation.

Batch: collection of independent requests to a model

significantly improve performance & reduce costs as the processing is done in parallel.

